

ADVANCED MICROECONOMETRICS
Problem Set - Day 1

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Total points: 100. Please return the problem set on time. Use your favorite software to answer the questions. Submit your problem set to: surzua@umd.edu

Q 1. Consider the following regression model:

$$Y_i = \alpha_0 + \alpha_1 X_i + V_i.$$

Assume the following parameterization of the model: $X \sim N(2, 2)$, $V \sim N(0, 4)$, $\alpha_0 = 2$, and $\alpha_1 = -1$.

- (a) Using this parameterization, generate a fake sample of 1,000 observations.
- (b) Code the associated likelihood function.
- (c) Estimate the parameters of the model using information on $\{Y_i, X_i\}_{i=1}^{1000}$ and your ML routine. Compare your results with the actual values.
- (d) Compute standard errors for your point estimates.

Q 2. Consider the following model:

$$D = 1[\alpha_0 + \alpha_1 Z > V].$$

and assume you observe $Y = a + bX + \epsilon$ if $D = 1$ and you observe $Y = 0$ otherwise.

Assume the following parameterization of the model: $Z \sim N(0, 2)$, $X \sim N(0, 1)$, $\epsilon \sim N(0, 1)$, $V \sim N(0, 1)$, $a = 0.5$, $b = -0.5$, $\alpha_0 = 2$, and $\alpha_1 = -1$.

- (a) Using this parameterization, generate a fake sample of 1,000 observations.
- (b) Code the likelihood function associated with this problem.
- (c) Estimate the parameters of the model using information on (Y, D, Z, X) and your ML routine. Compare your results with the actual values.
- (d) Compute standard errors for your estimates.

Q 3. Consider the following random utility model:

$$\begin{aligned} Y_1^* &= \beta_1 Z + V_1 \\ Y_2^* &= \beta_2 Z + V_2 \\ Y_3^* &= \beta_3 Z + V_3 \end{aligned}$$

and define $D = 1$ if $(Y_1^* > Y_2^*, Y_1^* > Y_3^*)$, $D = 2$ if $(Y_2^* > Y_1^*, Y_2^* > Y_3^*)$, and $D = 3$ otherwise, with

$$V \sim N \left[\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_{11}^2 & \sigma_{12} & \sigma_{13} \\ \sigma_{12} & \sigma_{22}^2 & \sigma_{23} \\ \sigma_{13} & \sigma_{23} & \sigma_{33}^2 \end{pmatrix} \right].$$

Consider alternative 3 as the benchmark, and assume the following parameterization: $Z \sim N(0, 3)$,

$$\begin{bmatrix} V_1 - V_3 \\ V_2 - V_3 \end{bmatrix} \sim N \left[\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & 0.5 \\ 0.5 & 2 \end{pmatrix} \right]$$

$\beta_1 = 0.25$, and $\beta_2 = 0.5$, $\beta_3 = 1$.

- Using this parameterization, generate a fake sample of 10,000 observations.
- Code the associated likelihood function.
- Show that you can identify $\beta_1 - \beta_3$, $\beta_2 - \beta_3$, $Cov(V_1 - V_3, V_2 - V_3)$, and $Var(V_2 - V_3)$.
- Estimate the parameters of the model using information on (D_1, D_2, D_3, Z) and your ML routine. Compare your results with their actual values. (Hint: Don't forget to impose the normalizations).

Q 4. Consider the following function:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left(-\frac{(x - \mu)^2}{2\sigma^2} \right)$$

and the fact that:

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

Answer the following questions:

- (a) Assuming $\mu = 1$, $\sigma^2 = 2$, use Gauss-Hermite quadrature with $n = 4$, $n = 6$ to compute the integral:

$$\int_{-\infty}^{\infty} x f(x) dx.$$

- (b) Assuming $\mu = 1$, $\sigma^2 = 2$, use Gauss-Hermite quadrature with $n = 4$, $n = 6$ to compute the integral:

$$\int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx.$$

(Hint: Change of variable).

Q 5. Consider the selection model discussed in Heckman 1974.

- (a) Describe the equations linking theory and practice. What are the key assumptions?
- (b) Download the NLSY97. Create a data set with the variables utilized by Heckman. Present descriptive statistics for your sample.
- (c) Using the NLSY97, estimate the model in Heckman 1974. (Discuss your optimization routine).
- (d) Estimate Heckman 1997 for different schooling levels (12 or less years of education versus more than 12 years of education)
- (e) Explain under what assumption the average observed wage would be larger than the unconditional average wage.